

Chapter 8: Generative Adversarial Network

Generative Adversarial Network: Understand the process of alternatively updating discriminator and generator in adversarial learning.

Generative Adversarial Network: Derive that, under some conditions, the adversarial learning minimizes the Jensen-Shannon divergence between data and model distributions.

Representation Learning with Adversarial Learning: Understand how BiGAN introduces encoder to learn representations.

Representation Learning with Adversarial Learning: Derive that, BiGAN also minimizes the Jensen-Shannon divergence between data and model distributions.

Data-to-Data Translation: Understand the difference between paired and unpaired datasets.

Data-to-Data Translation: Understand how the CycleGAN learns the translator without pair information.

Extension to f -divergence: Show that both VAEs and GANs can be interpreted as introducing auxiliary networks, encoder and discriminator, to approximate the f -divergence between data and model distributions.

Extension to f -divergence: Understand that all generative models reviewed until Module 2 utilize likelihood ratio-based statistical distances.